Numerical Simulation and Obtaining Models from Experimental Data

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1 Introduction

- Numerical simulation refers to simulating the differential equations of a dynamic system on a digital computer (possibly nonlinear, including the controller).

- System identification (System ID) refers to determining the system model from experimental data.

In this lecture we will consider:

1. Simulation (briefly)

   (a) Simulation by Analog Computer

   (b) Simulation by Digital Computer

2. System Identification

   (a) By step response

   (b) By frequency response

3. Conclusions

2 Simulation

- Numerical simulation involves integrating differential equations to determine the *specific* system output for a *specific* input.

Enrichment Topic.

Pros: – Consider full (possibly nonlinear) system, no approximations.

– Works when no other analysis tools are available.

Cons: – Output applies only to that *specific* instance:

* The particular parameters used;
* The particular inputs used.

2.1 Method of Simulation:

- Given:

$$\dot{x}(t) = f(x, u) \tag{1}$$

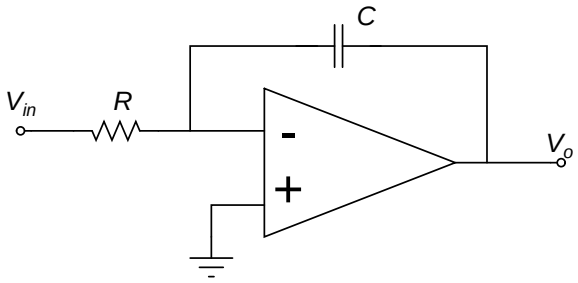
then

$$x(t) = \int_0^t \dot{x}(\tau)\, d\tau = \int_0^t f(x, u)\, d\tau \tag{2}$$

- Equations such as (2) can be integrated on either analog or digital computers.

2.2 Analog computers

- Op Amps were invented to build Analog computers.
- Analog computers integrate differential equations using capacitors as integrators:



$$V_o(t) = -\int_{\tau=0}^{t} \frac{1}{RC} V_{in}(\tau)\, d\tau \tag{3}$$

Figure 1: Op Amp Integrator.

- The output, $V_o(t)$, can be fed back through possibly nonlinear circuits to implement $f(x, u)$.
- Analog computers were used until recently for high-order simulations where many many simulations must be done, because Op Amps can run at MHz.
  - # Integrators = Model Order = # State Variables.

2.3 Numerical Simulation (numerical: by digital computer)

- Numerical simulation runs by computing Δ_x in small steps.

$$x(t_{i+1}) = x(t_i) + \Delta_x(t_i) \tag{4}$$

  - Where Δ_x can be computed by various methods, with various degrees of complexity and accuracy.
  - Matlab provides very good numerical integration tools.
  - Δ_x is the area under the curve in figure 2, Euler's method and the trapezoidal rule are two numerical integration methods.
  - Euler's method (dark area in figure 2) is the simplest:

$$\Delta_x(t_i) = f(x(t_i), u(t_i))\ \Delta_t \tag{5}$$

    or

$$x(t_{i+1}) = x(t_i) + \dot{x}(t_i)\,\Delta t \tag{6}$$

    where $\Delta_t = t_{i+1} - t_i$.

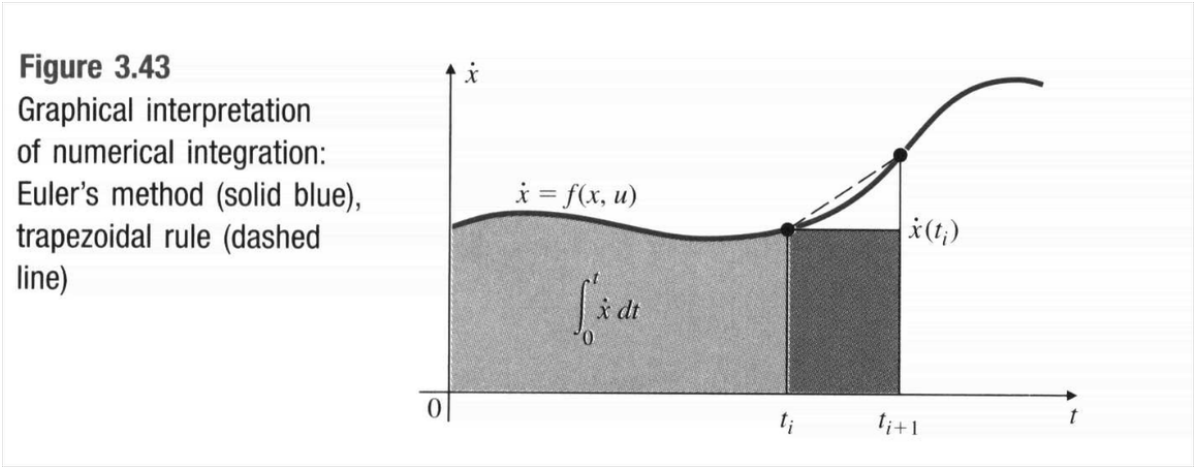


Figure 2: Numerical (digital computer) integration of differential equations.

Numerical Simulation (continued)

- The biggest digital computers have always been built to perform numerical simulations:
  - For years, each successive generation of Cray was used to simulate nuclear explosions (note: export of the biggest computers is controlled by the U.S. government).
  - The biggest computers today are often used to simulate the biosphere, to study global warming.
- Matlab functions such as `lsim()` permit simulation of linear systems for arbitrary inputs.
  - For linear systems with suitable inputs, Δ_x can be computed exactly.
- The Matlab `step` function does not use Numerical Simulation.
  - In the `step` function, Matlab solves the inverse Laplace transform problem (using the cover-up method).

2.4 Conclusion

- Simulink is powerful and widely used.
- The need for analog simulation lead to the invention of the Op Amp.
- Numerical simulation is widely used in controls work today.

3 System Identification (Franklin *et al.* section 3.7 (7th ed.))

3.1 Introduction

- System ID is the determination of a system model from experimental data.

3.1.1 System ID comprises two parts

- System identification comprises two inter-related portions:

  1. Determining the model structure (number of poles, presence of pure delay, etc.)

  2. Determining the values of parameters within the model structure.

- For example, in Lab Part 3, you determined model parameters, but the model structure was given.

- System ID is a key topic, because the controls designer rarely knows all the needed parameters from equipment data sheets.

- All controls engineers encounter System ID problems.

- In practice, System ID is expensive. It is a barrier to good control.

  – $100M+ can be spent building the dynamic models for flight controls of military aircraft.

  – For manufacturing equipment or building heating, the models are usually simpler, but System ID is still a big cost.

Introduction: System ID (continued)

- There is often considerable room for improvement in the efficiency of building environmental control systems (HVAC: Heating Ventilation and Air Conditioning) because:
  - The thermal model of the building is complex, interconnected and poorly known;
    - As a result, the control can be poorly tuned.
- Adaptive control seeks to automatically identify the system parameters, and automatically tune the controller (thus less manual System ID).
  - Adaptivecontrol has been investigated since the early 1960's.
  - Since about 1990, the number of practical applications is growing.
    - Numerous adaptive controllers now on the market.
    - Most automobile engine controllers now have some adaptation going on (e.g,. to changing air pressure or humidity).
    - Adaptive ship autopilots.
  - Because of the breadth of:
    - System characteristics,
    - Operating conditions, and
    - Idiosyncrasies,

    We are still far from the day of the one-size-fits-all adaptive controller.

Introduction (continued)

3.1.2 Approaches to System ID

- For our purposes, methods of System Identification can be classified into 3 groups:

  1. Identification from the step response;
  2. Identification from the frequency response;
  3. Advanced methods (not covered here).

3.1.3 Pure delay

- Some systems have pure delay, also called 'transport lag' (illustrated in figure 3, below). For example: A new model element

  – Automobile engine model, there is a pure delay between a change in the fuel injection command and a change in the torque;

  – Due to the time required for the compression stroke.

- Modeling pure delay (a new linear model element):

  – Pure delay has an impulse response of:

$$h(t) = \delta(t - \tau_d) \tag{7}$$

  – With a transfer function of:

$$G_d(s) = e^{-s\tau_d} \tag{8}$$

  Note that $G_d(s)$ is not a rational polynomial. The Padé approximation gives a rational-polynomial approximation to Eqn (8).

3.2 System ID with step input

- Several distinct quantities that can be measured from step response are illustrated in figure 3.

1. DC Gain K_{DC}

   DC gain is an essential measurement from the step response. Note that DC gain can is infinite for some systems (in which case, typically, no measurements can be made from step response).

2. Delay τ_d

   Some systems have pure delay between the input and the beginning of the output. A fluid flowing through a pipe (say, between a valve and a tank) gives pure delay.

3. Rise Time τ_r, τ_p or τ_{ZL}

   We have already seen that rise time can be defined more than one way. Here the 10-90 rise time, and the 'Ziegler-Nichols' rise time are defined.

4. Overshoot Magnitude M_p

   The peak time is related to the rise time. M_p is an independent measurement.

5. Settling Time τ_s

   For this case, $\tau_s = \tau_d + (4/\alpha)$.

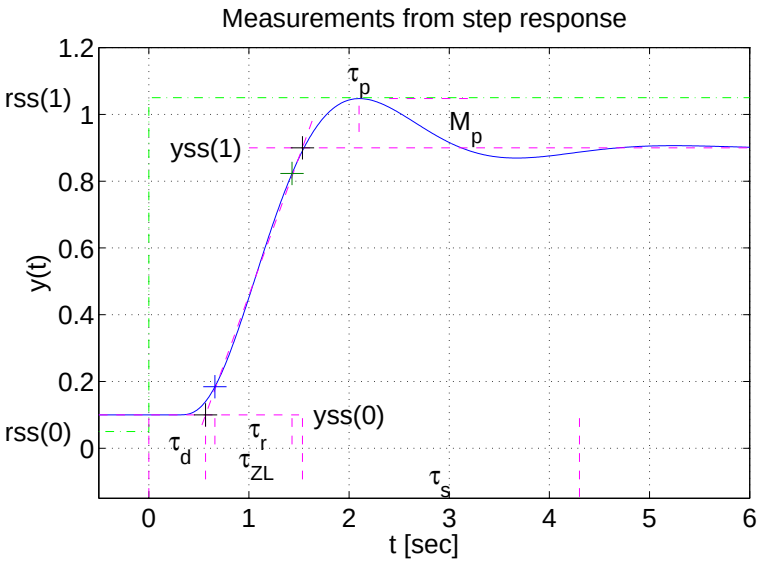


Figure 3: Quantities measured in step response for System ID.

System ID with step input (continued)

- Details of determining model parameters from the measurements depend upon the model structure.

- The step response can be useful for determining characteristics of the dominant mode,

  – Because the dominant mode dominants the response.

- For a 2nd order dominant mode, one could use:

$$G_p(s) = K_{DC} \frac{\omega_n^2}{s^2 + 2\alpha s + \omega_n^2} \tag{9}$$

where

1. K_{DC} can be estimated from Δ_y/Δ_r;
2. The decay rate, α, can be estimated from τ_s; and
3. Natural frequency, ω_n, can be estimated from the damping factor.
   - If we have overshoot, estimate the damping factor from the overshoot M_p, and
   $$\rho \simeq \sqrt{\frac{\ln(M_p/100)^2}{\pi^2 + \ln(M_p/100)^2}}$$
   - Then
   $$\omega_n \simeq \frac{\alpha}{\rho}$$
4. The damped natural frequency ω and pole locations can be determined from ρ and ω_n.

3.3 System ID from frequency response

- The dominant mode dominates the step response.
  - Using the step response, it is difficult or impossible to identify modes beyond the dominant mode.
- By applying sine waves at various frequencies, we can measure the Magnitude and Phase of the response at any practical frequency, ω_i.
  - 2 data points per ω_i ... possible to have many independent data points;
  - Testing at a higher frequency tests response of modes near that frequency, not only the dominant mode.

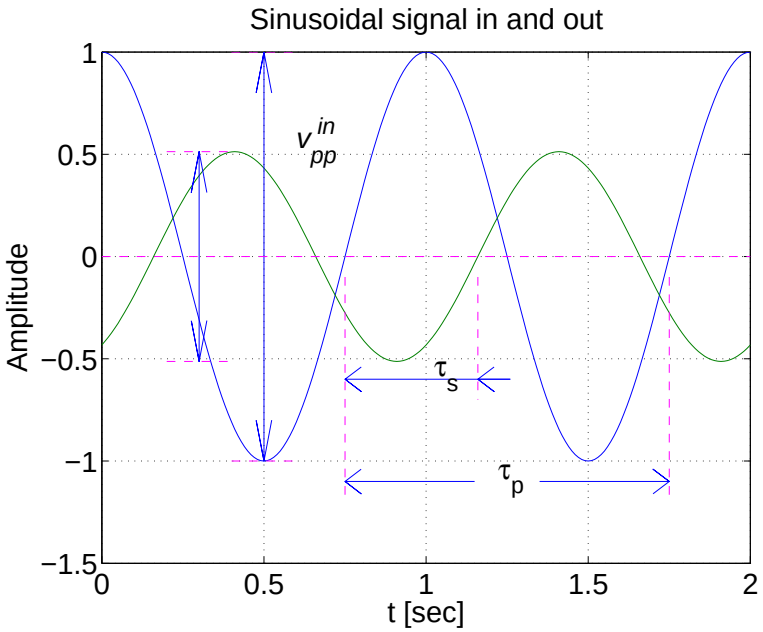


Figure 4: Illustration of complex gain.

- Using the measurements in figure 4, the complex gain, $M \angle \phi$, is given as:

$$M = v_{pp}^{out} / v_{pp}^{in} \tag{10}$$

$$\phi = -360 \frac{\tau_s}{\tau_p} \text{ [deg]} \tag{11}$$

System ID from frequency response (continued)

- Sometimes higher-frequency modes can be important, such as a flexible mode of a mechanism that might introduce stability problems.

  Example: Web Transport.

  - A The rollers, etc. that move paper along in paper making are called the "web transport."

  - A drive for web transport can have a long shaft, with a lot of rotating mass at the end. This gives a:

    Lightly-damped, flexible mode between the motor and the roller.

  - The dominant mode may be a first-order response of the velocity, but knowing the locations of the resonant mode poles is essential for good design.

3.3.1 System ID from frequency response: Method

- Testing comprises applying sinusoid signals at various frequencies,
  - Appropriate frequencies and amplitudes must be selected.
- The experiment may be conducted either open loop or closed loop;
  - Open-loop methods give more direct measurement of the system model.
  - Closed-loop methods can be used while the plant is running.

3.3.2 System ID from frequency response: Steps

1. Select appropriate frequencies and amplitudes of the input signal.
2. Measure $H(j\omega)$, the magnitude and phase of the output with respect to the input, for each input:
$$H(j\omega)\Big|_{\omega=\omega_i} = \frac{\vec{y}}{\vec{u}} \tag{12}$$
3. Select the model structure (number of poles, n_a, and zeros, n_b).
4. Adjust the model parameters so that the model best fits the observed frequency response (using Matlab function `invfreqs()`).

3.3.3 System ID from frequency response: Example

- Consider a velocity motor servo with the transfer function:

$$H(s) = K_{DC}\ \frac{b}{Js+b} \tag{13}$$

where $K_{DC} = 6\ \left[\frac{\text{rad/sec}}{\text{volt}}\right]$, $J = 1.75\ \left[\text{Kg}-\text{m}^2\right]$ and $b = 7\ \left[\frac{N-m}{\text{rad/sec}}\right]$.

- The bode plot is shown in figure 5, along with example measured data. Note: the measured data includes experimental noise, noise = 10% of signal.

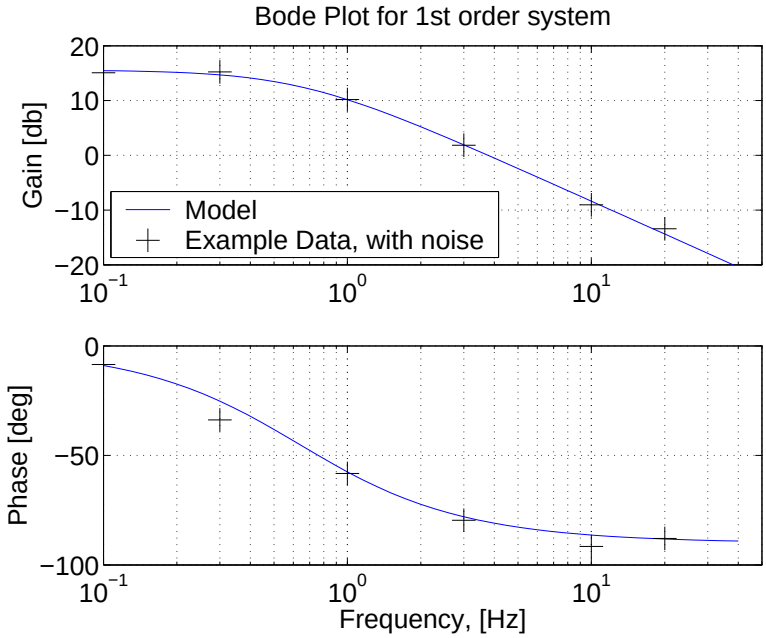


Figure 5: Bode plot of motor servo, with noisy experimental data.

1. Select appropriate frequencies:
   - For accurate identification, test points should cover in the region where the response changes: this suggests 0.5 ... 40 [rad/sec].
   - Test points should also span the frequency range of interest, for example, if it is known that response up to 20 [Hz] is important, the coverage of test points should be extended up to this frequency.
   - Possible selection of frequencies (seen in Fig 5):

$$F_s:\ \{0.1, 0.3, 1.0, 3.0, 10.0, 20.0\} \quad [\text{Hz}]$$

Example (continued)

- The frequencies most useful for identifying the dynamic model are those where the bode plot changes, especially the phase.

- If the frequencies of the dynamics of interest are not known in advance, the frequency response can be explored.

- An example data set for $G_p(s)$ is seen in table 1.

Frequency		Gain		Phase	$\overline{H}(j\omega)$
[Hz]	[rad/sec]	[Magnitude]	[dB]	[deg]	Cmplx Gain
0.10	0.63	6.35	16.06	-12.73	6.198 - j 1.400
0.30	1.88	5.50	14.80	-24.89	4.988 - j 2.314
1.00	6.28	3.36	10.54	-51.22	2.107 - j 2.623
3.00	18.85	1.28	2.14	-83.25	0.151 - j 1.271
10.00	62.83	0.44	-7.04	-87.50	0.019 - j 0.444
20.00	125.66	0.19	-14.37	-88.62	0.005 - j 0.191

Table 1: Measured data for system of Eqn (9).

- `invfreqs()` runs with the gain as a single complex number, as shown in the last column.

$$\overline{H}(j\omega) = M(j\omega)\angle\phi(j\omega) \tag{14}$$

where $M(j\omega)$ is the magnitude, column 3, and $\phi(j\omega)$ is the phase, column 5.

3.3.4 Matlab function invfreqs()

- Determining the best fit of poles and zeros from measured response data is a non-linear parameter estimation problem.

  – It operates by minimizing a "cost" function of the form

$$J = \sum_{k=1}^{N_w} \left(\left| \widehat{H}_k - \overline{H}_k \right| W_t^{(k)} \right)^2 \tag{15}$$

  where $\overline{H}_k$ is the kth sample of measured data, $\widehat{H}_k$ is the k_{th} response predicted from the model, and $W_t^{(k)}$ is a weight applied to the kth sample.

  – Non-linear parameter estimation is beyond our scope.

  – This non-linear fit is done by the Matlab routine `invfreqs()`.

- Read

```
>> help freqs
>> help invfreqs
```

3.3.4.1 How `invfreqs()` works

- As with many things, it is necessary to know something of how the `invfreqs()` function works in order to use it successfully. The call is:

  `[B, A] = invfreqs(`$\overline{H}$`, W, `n_b`, `n_a`, `W_t`)`

- The call to `invfreqs()` is shown above. The arguments are:

  $B(s)$: Numerator polynomial vector, $B \in R^{1\times(n_b+1)}$.

  $A(s)$: Denominator polynomial vector, $A \in R^{1\times(n_a+1)}$.

  $\overline{H} \in R^{N_\omega\times 1}$: A vector of measured complex gain values; one gain value for each frequency in `W` (e.g., table 1).

  $W \in R^{N_\omega\times 1}$: The frequencies [rad/sec] at which gains H were measured.

  N_ω: The number of data points.

  n_b: The number of zeros to put in the model.

  n_a: The number of poles to put in the model.

  $W_t \in R^{N_{omega}\times 1}$: Optional vector of weights, used to compute the error function. Typical choice: $W_t = 1./\operatorname{abs}\left(\overline{H}\right)$.

3.3.4.2 The invfreqs() Process:

1. Determine an initial estimate for the transfer function:

$$\frac{\widehat{B}(s)}{\widehat{A}(s)} = \widehat{G_p}(s) \tag{16}$$

2. Begin loop

   (a) The complex gain given by the current estimate for $\widehat{T}(s)$ is calculated at frequencies W:

$$\widehat{H} = \text{freqs}(\widehat{B}, \widehat{A}, W) \tag{17}$$

   (b) The goodness of the fit is measured with the sum-squared-error:

$$J = \sum_{k=1}^{N_w} \left(\left| \widehat{H}_k - \overline{H}_k \right| W_t^{(k)} \right)^2 \tag{18}$$

   Note:

   - $\overline{H}$ indicates measured data,
   - $\widehat{H}$ indicates estimated data, based on the model, and
   - H^* indicates true or perfect data (generally unknown).

   (c) Adjust the parameters, $\widehat{B}$ and $\widehat{A}$, to reduce errors.

   (d) Repeat loop until J can not be further reduced.

3.3.4.3 Some things `invfreqs()` does not do

1. `invfreqs()` does not determine how many poles and zeros are present in the system (the model structure), the user must specify n_a and n_b.
2. `invfreqs()` does not determine if the model produced is physically meaningful.

- Items 1 and 2 must be considered by the user.

3.3.4.4 Using invfreqs()

- The steps listed in section 3.3.2 are:

  1. Select appropriate frequencies and amplitudes of the input signal.
     - Discussed above.
  2. Measure $\overline{H}(j\omega)$ at each ω.
     - Procedure depends on equipment.
  3. Select the model structure (n_a and n_b).
     - This the user must do.
  4. Adjust the model parameters for best fit to model data.
     - This is done by `invfreqs()`.

- Steps 1, 2 and 4 have been addressed, only 3 remains.

3.3.4.5 Selecting the model structure

- The model structure may be known from first principles.
- Otherwise:
  - It may be necessary to test several models (several combinations of poles and zeros) and select the most suitable one.
- Some ideas to keep in mind are:
  - When `invfreqs()` gives an estimated model, we can evaluate J with Eqn (18).
  - The best fit (smallest J) may not be the most suitable model:
    - A simple model is preferred over a complex one.
    - As the model becomes more complex (more poles and zeros)
      - The value of J will always go down, just because there are more poles and zeros to adjust;
      - But the model DOF may be extra and `invfreqs()` may be fitting parameters to noise.

Selecting the model structure (continued)

- Common-sense notions are also important:

  1. If the open-loop system is stable, and the model has unstable modes $\rightarrow$ reject the model.

  2. If the data set lies on the range $\omega_{\min} \cdots \omega_{\max}$,

     – Poles or zeros very far from this range can not be accurately identified.

     – If $|p_i|$ or $|z_i| << \omega_{\min}$; or $|p_i|$ or $|z_i| >> \omega_{\max} \rightarrow$ reject the model, and choose a simpler model.

  3. Right-half plane zeros (also called non-minimum phase zeros) arise only in special cases. Non-min Phase zero

     – If the open-loop system is known to be without right-half plane (RHP) zeros, and the model includes one or more RHP zeros $\rightarrow$ reject the model.

  4. If the experiment is repeated, the same model should be produce.

     – If the experiment is repeated and the poles and zeros change significantly $\rightarrow$ reject the model.

Finally

- If more than one model satisfies the common-sense criteria above,

  – Select the model with the lowest J and which has the least parameters.

  – Require a reasonable improvement in J (such as a 25% improvement) before accepting additional parameters. (The Fisher statistic gives a rigorous test for the improvement)

Summary

- Two models being nearly equal, the simpler one is preferred.

3.3.5 Example, Best Fit

- Terms $H^*(s)$, $\widehat{H}(s)$ and $\overline{H}(j\omega)$ (the true model, the estimated model and the data) are seen in figure 6.

$H^*(s)$, $\widehat{H}(s)$, $\overline{H}(j\omega)$

- $H^*(s)$, the true model (generally unknown) is:

$$H^*(s) = K_{DC}\frac{b}{Js+b} = 6.00\frac{7.00}{1.75s+7.00}$$

$$K_{DC}{}^* = 6.00 \quad ; \qquad \tau^* = 4.00\ [\text{sec}]$$

- $\widehat{H}(s)$, the model estimated from the data is:

$$\widehat{H}(s) = K_{DC}\frac{b/J}{s+b/J} = 6.50\frac{3.82}{1.0s+3.82}\ ; \tag{19}$$

$$\widehat{K_{DC}} = 6.50 \quad ; \qquad \widehat{\tau} = 3.82\ [\text{sec}]$$

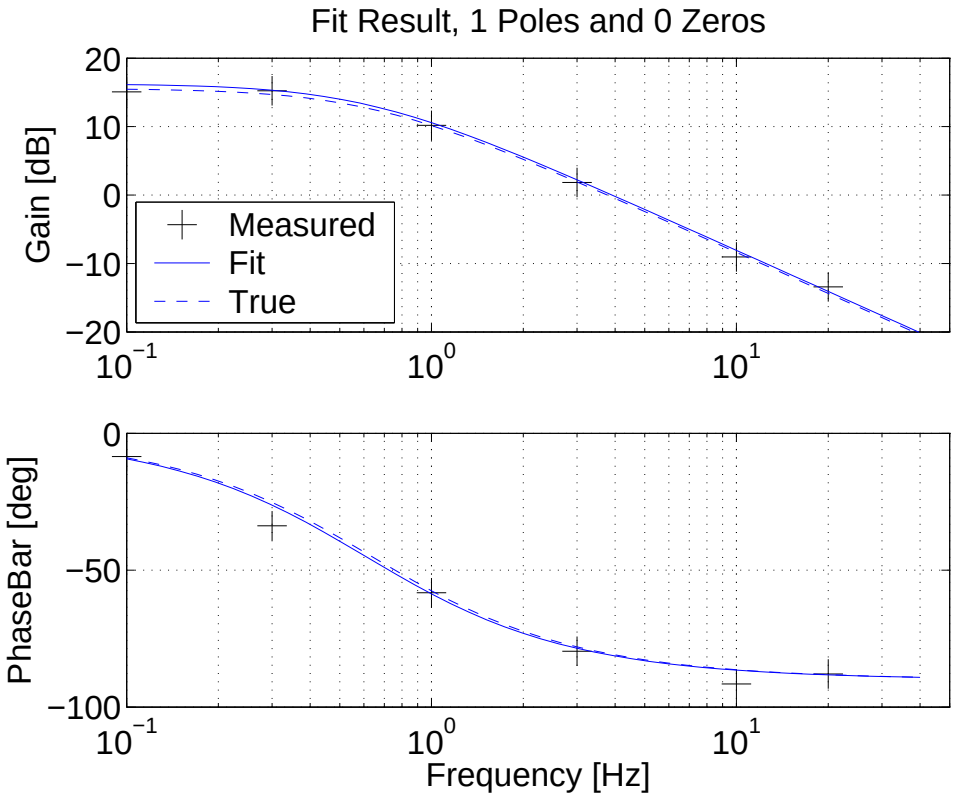


Figure 6: Bode plot of $H^*(s)$, $\widehat{H}(s)$ and $\overline{H}(j\omega)$.

System ID may not give the physical parameters

- `Invfreqs()` always returns $a_0 = 1.0$. Experimental parameter determination (by any method) is not able to distinguish:

$$H(s) = 6.0\frac{7.0}{1.75s+7.0} \qquad \text{from} \qquad H(s) = 6.0\frac{4.0}{1.0s+4.0}$$

- For example, with

$$G_{uy}(s) = \frac{k_t}{Js+b}$$

to determine the actual, physical values of J and b, at least one additional piece of information is required.

3.3.6 Example of model selection

- Using the example above, information needed to select model structure is seen in table 2.
  - 9 models are tested with `invfreqs()`.
  - Models with 1, 2, 3 or 4 poles, and with 0, 1 or 2 zeros, are considered.

Trial	Number of Poles	Number of Zeros	J	ω_n Fastest Pole(s)	Unstable Pole	ω_n Fastest Zero	RHP Zero	DC Gain
1	1	0	0.2172	3.82		-	-	6.50
2	2	0	2.1008	112.77		-	-	3.82
3	2	1	1.5540	7.69	X	5.37	X	3.82
4	3	0	0.1835	247.06		-		6.43
5	3	1	0.1629	251.61	X	147.04	X	6.26
6	3	2	0.1527	75.82	X	73.28	X	6.26
7	4	0	0.1345	179.27	X	-		6.02
8	4	1	0.1581	240.13	X	45.27		6.02
9	4	2	0.7664	248.06	X	149.15	X	4.70

Table 2: Results of fitting various models to the data of table 1.

- In this case, model 1 is the only valid choice:
  - Models 3, 5, 6, 7, 8, 9 are unstable;
  - Model 2, 3 and 9 have a much higher J ;
  - Models 4, 5, 8 and 9 have modes or zeros that are much faster than the data range, 0.63 .. 125 [rad/sec].

Example of model selection (continued)

- The Bode plot corresponding to trial 4 is seen in figure 7.
  - The model fits quite well across the range of the data, but the extra complex mode does not correspond to anything in the data.
  - Looking at table 2 shows that trail 4 gives a mode at $\omega_n = 247$ [rad/sec], whereas the data ends at 125 [rad/sec] .

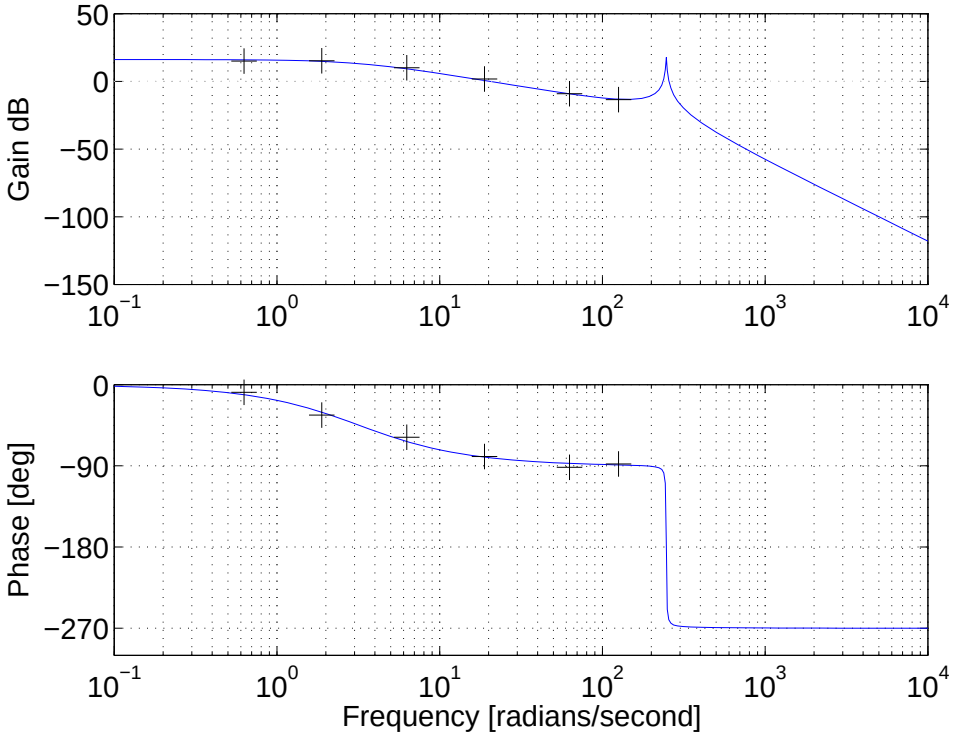


Figure 7: Bode plot of case 4, showing $H^*(s)$, $\widehat{H}(s)$ and $\overline{H}(j\omega)$.

Example of model selection (continued)

- Another view of the model selection process is seen in figure 8.
- The pole-zero constellations are all over the map.
  - This high variability arises with model parameters fitting noise.
- For the most part, these trials can be rejected just by looking for unstable modes and the scale of the axes of the S-plane plot.

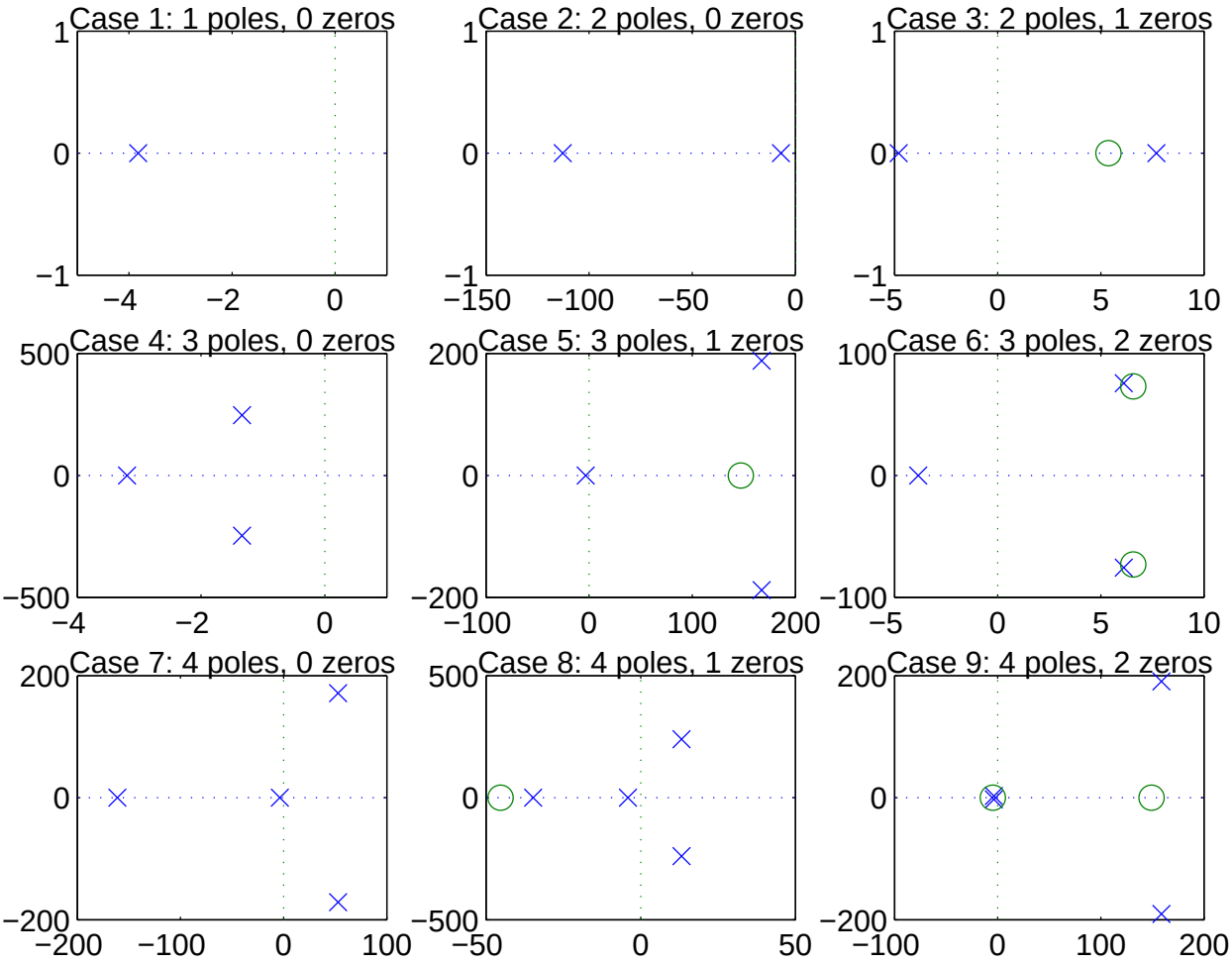


Figure 8: Pole zero diagrams for 9 different model structures.

Example of model selection (continued)

- The 9-model experiment is repeated with a different sample of noise
  - Same %10 noise level as in figure 8, different random noise values.

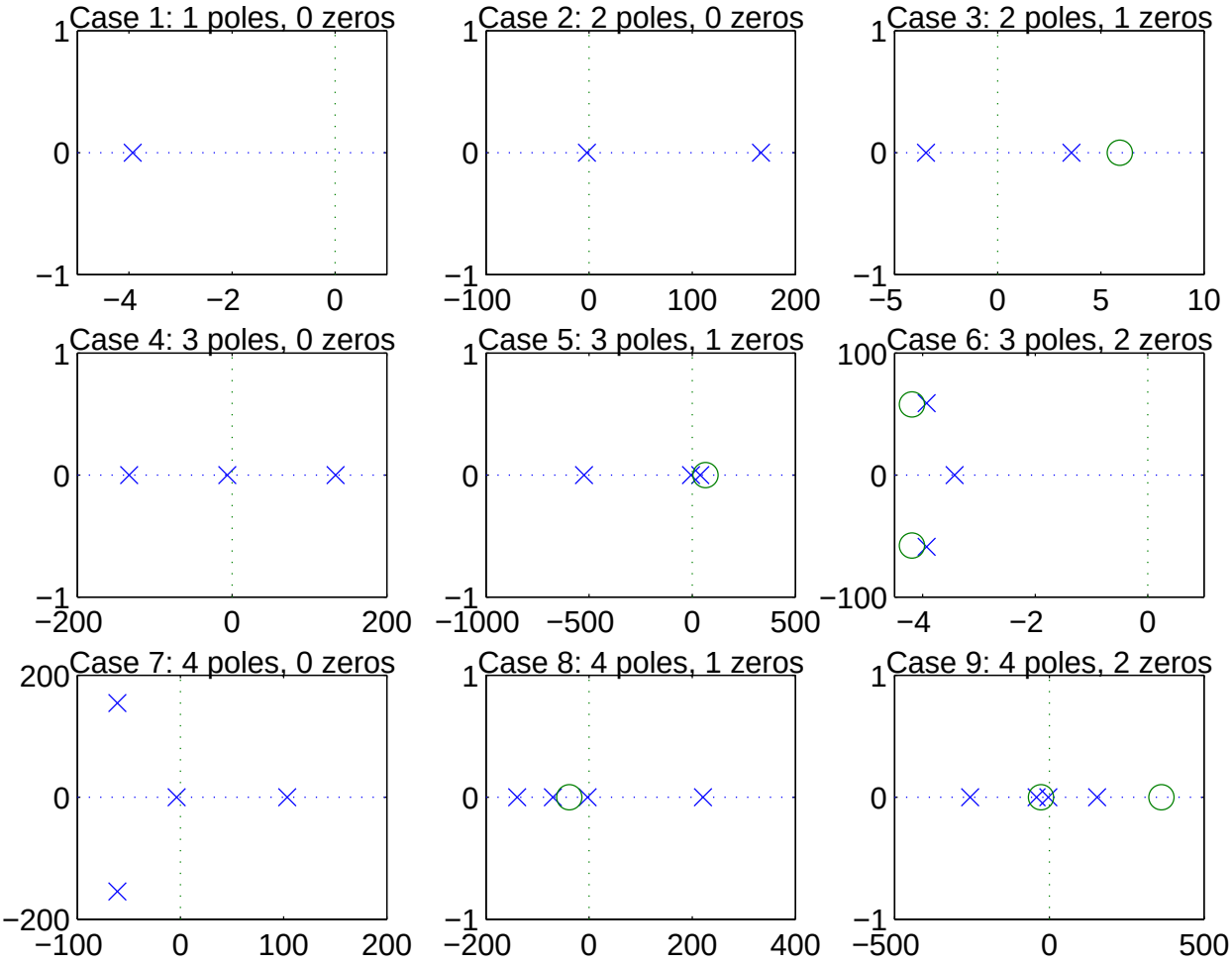


Figure 9: Pole zero diagrams, repeated experiment with different sample of noise.

- Comparing figures 8 and 9, one sees that the extra poles and zeros move around quite a lot, in response to a new sample of noise.
- This was mentioned above, as a way to test if model poles or zeros are real, or just fabrication of the model fitting process.

3.3.7 Second Example of Model Selection

- A second example also illustrates the model determination process.
- The hypothetical system is 3rd order with one zero:

$$H^*(s) = 6.0\,\frac{11.9\,s + 119.0}{1.75\,s^3 + 10.50\,s^2 + 43.75\,s + 119.0} \tag{20}$$

- For this case, noise has been reduced to 5%
  - The greater the number of parameters, the more sensitive identification becomes to noise.
- The pole-zero map is seen in figure 10; which shows both:
  - The ideal pole locations, from $H^*(s)$;
  - The identified pole locations, from $\widehat{H}(s)$.

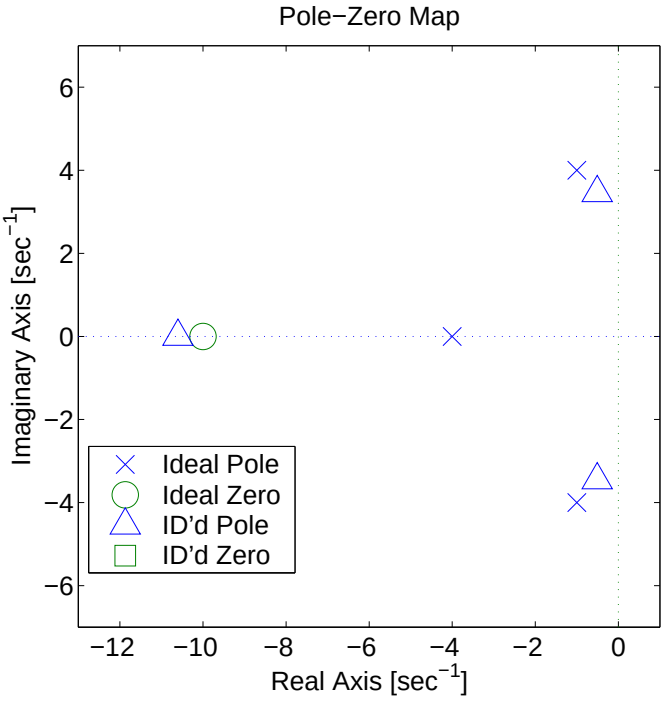


Figure 10: P-Z map for 3rd order example.

Second Example of Model Selection (continued)

- The data and results are illustrated in figure 11.
- The identified model is:

$$\widehat{H}(s) = 6.24 \frac{6.74s + 129.58}{s^3 + 11.62s + 23.05s + 129.58} \tag{21}$$

- Figure 11 shows that the identified model, along with the ideal and the data.

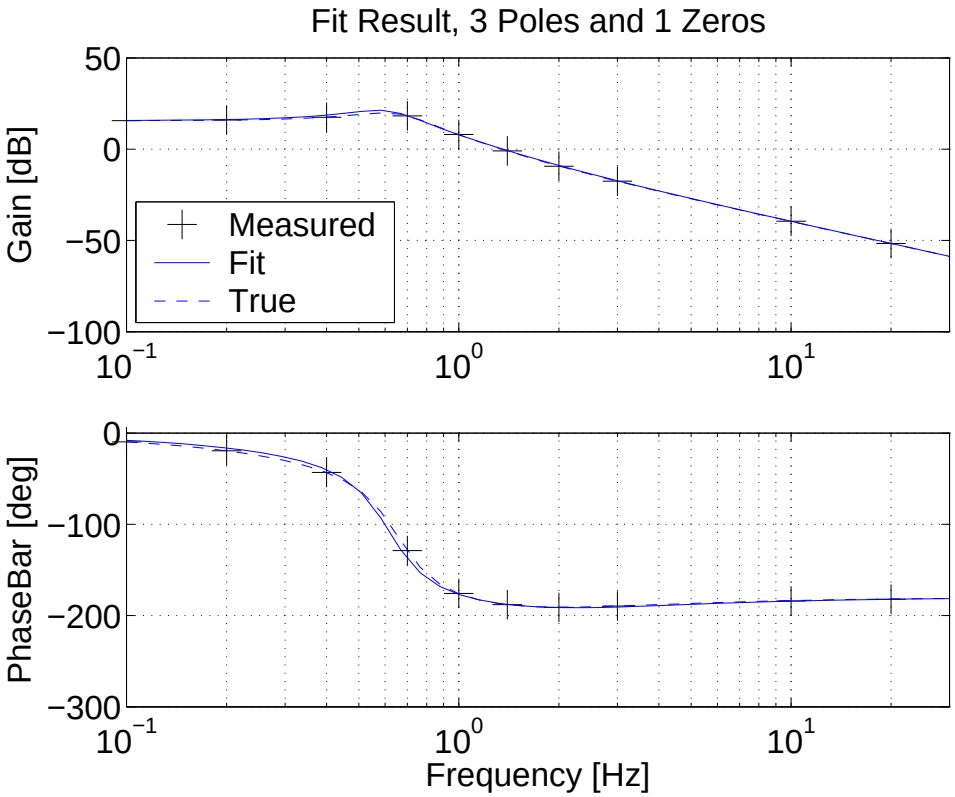


Figure 11: Bode plot of $H^*(s)$, $\widehat{H}(s)$ and $\overline{H}(j\omega)$ for 3rd order example.

Second Example of Model Selection (continued)

- Table 3 shows the numerical results.

Trial	Number of Poles	Number of Zeros	J	ω_n Fastest Pole(s)	Unstable Pole	ω_n Fastest Zero	RHP Zero	DC Gain
1	1.00	0.00	31.7811	0.0038		-		-8.12
2	2.00	0.00	22.2490	3.6	X	-		3.34
3	2.00	1.00	14.9818	3.6	X	1.9e+03	X	3.34
4	3.00	0.00	7.1464	9.3e+02		-		3.37
5	3.00	1.00	5.7894	11		19		6.24
6	3.00	2.00	2.0232	7.2		7.4e+03	X	6.24
7	4.00	0.00	30.5442	1.6e+02	X	-		6.27
8	4.00	1.00	1.0449	5.6e+03		13		6.27
9	4.00	2.00	10.8100	33	X	39	X	2.12

Table 3: Results of fitting various models to the data of the 3rd order example.

- Eight out of Nine trials can be rejected:
  - Trials 2, 3, 7 and 9 can be rejected because they have unstable modes.
  - Trials 1, 2, 3 and 7 can be rejected because they have high residual errors.
  - Trials 4 and 8 can be rejected because the fitting process produced modes much faster than the measured data.
  - Trials 3 and 6 can be rejected because the fitting process produced zeros much faster than the measured data.
- The remaining trial is the correct one.

Second Example of Model Selection (continued)

- The pole-zero constellation of each trial is shown in figure 12.

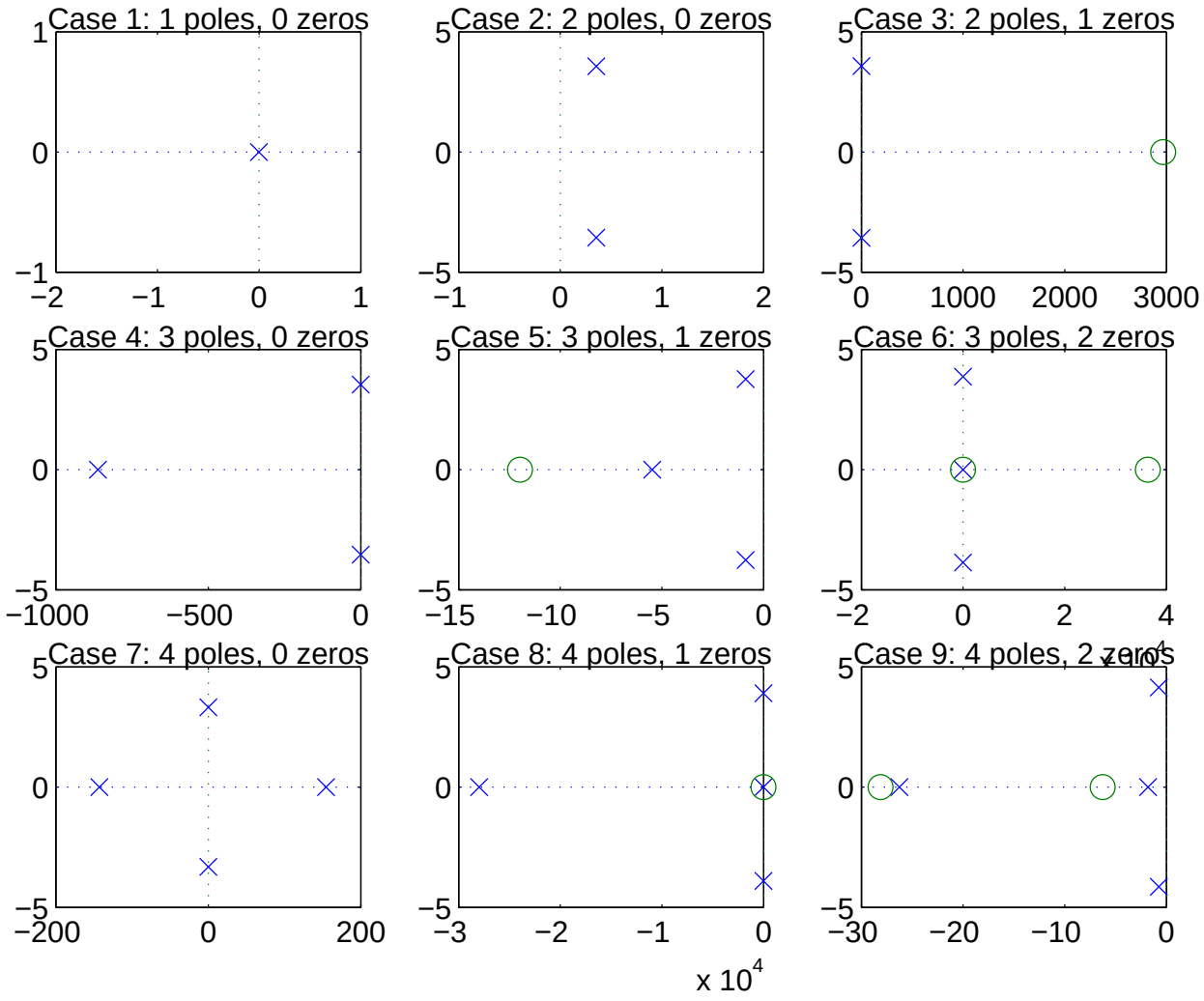


Figure 12: Pole zero diagrams for 9 different model structures, 3rd order example, data of table 3.

3.3.8 Using a priori information with System ID

- Leonard Lung[1] once called System ID 'a fiddler's paradise,' because each situation has its own *a priori* [2] information that needs to be considered.
- Information the invfreqs() algorithm explicitly uses:
  - $\overline{H}(j\omega)$ data as given.
- Sources of *a priori* information:
  - The model structure may be known,
    - For example, from first-principles analysis of the system.
  - DC Gain may be known,
    - For example, from a step-input test.
  - Some poles and/or zeros of the system may be known,
    - For example from first-principles, or
    - From one well-documented element in series with the rest.

1. Co-Author of "Theory and Practice of Recursive Identification," and many papers on System ID.
2. *a priori*: known before hand.

3.3.8.1 If model structure is known

- Use the known model structure, do not test many models and pick the best.

3.3.8.2 If DC Gain is known

- The data set of table 2 or figure 6 shows gain and phase over a range of frequencies, but not at DC.
- The algorithm was not fitting any DC information, the result is a less-than-ideal fit to DC gain (6.50 rather than 6.00).
  - If the DC gain is known *a priori* , put it in as an additional data point:

$$\omega_1 = \left[0 \ ; \ \omega_0 \right]; \qquad \overline{H}_1 = \left[\overline{K_{DC}} \angle 0; \ \overline{H}_0 \right]; \qquad W_t^1 = [\ \text{Large}; \ W_t^0\ ]$$

Frequencies, Complex gain, Weights

where

  - ω_0 is the initial list of freqs;
  - $\overline{H}_0$ is the initial measured values for complex gain;
  - W_t^0 is the initial weights, the more accurately the DC gain is known, the larger the corresponding weight should be.

Comparison of results, with DC Gain used

- Running `invfreqs()` with W_1, $\overline{H}_1$, and W_t^1 gives:

$$\widehat{H}(s) = 6.02 \frac{1}{s+4.12} \tag{22}$$

- System ID results are compared in table 4.
  - Incorporating *a priori* knowledge of the DC gain improves the estimated model accuracy.

	Pole location	DC Gain (3)
True	$s = -4.0$	6.0
Full Model Estimation (sect. 3.3.5)	$s = -3.82$	6.50
Using known DC Gain	$s = -4.12$	6.02

Table 4: Comparison of characteristics of $H^*(s)$ with estimate $\widehat{H}(s)$, with and without using prior knowledge of the DC gain.

3.3.8.3 If Known Dynamics are present:

- Sometimes a system model can be broken down into:

$$H(s) = H^0(s)\, H^1(s) \tag{23}$$

  where:

  - $H^0(s)$ is an *a priori* known portion of the dynamics, and
  - $H^1(s)$ must be determined.

For example, when the motor servo output is position, $G_p(s)$ is known to have a pole at the origin (because position is the pure integral of velocity, $H^0(s) = \frac{1}{s}$).

- In such a case, only the unknown dynamics need be estimated:

  1. Removing the influence of the known dynamics from the measured data:

$$\overline{H}^1(j\omega) = \frac{1}{H^0(j\omega)}\,\overline{H}(j\omega) \tag{24}$$

     where:

     - $\overline{H}(j\omega)$ is the measured data;
     - $1/H^0(j\omega)$ is the reciprocal of the contribution of $H^0(j\omega)$;
     - $\overline{H}^1(j\omega)$ is the data which would have been measured on system $H^1(s)$.

  2. Next, use data $\overline{H}^1(j\omega)$ to estimate $\widehat{H}^1(s)$ with invfreqs (in the usual way).

  3. Finally $\widehat{H}(s)$ is given by:

$$\widehat{H}(s) = H^0(s)\,\widehat{H}^1(s) \tag{25}$$

Known Dynamics (example)

- For example, consider:

$$H^*(s) = \frac{24}{(s+0)(s+4.0)} \tag{26}$$

  which is the first example with an added pole at the origin (position output).

- Example data with 5% noise are given in table 5.

W [rad/sec]	$\overline{H}(j\omega)$	$H^0(j\omega)$	$\overline{H}^1(j\omega)$
0.63	9.40 < -100.2	1.59 < -90.0	5.91 < -10.2
1.88	3.11 < -113.8	0.53 < -90.0	5.86 < -23.8
6.28	0.52 < -147.5	0.16 < -90.0	3.24 < -57.5
18.85	0.07 < -168.3	0.05 < -90.0	1.26 < -78.3
62.83	0.01 < -178.5	0.02 < -90.0	0.40 < -88.5
125.66	0.00 < -174.8	0.01 < -90.0	0.19 < -84.8

Table 5: Data showing removal of pole at origin.

- If the 2nd order dynamics are estimated straight from $\overline{H}(j\omega)$, the result is:

$$\widehat{H}(s) = \frac{24.27}{(s+0.023)(s+5.059)} \tag{27}$$

  Neither the DC Gain nor the pole locations are very accurately estimated.

Known Dynamics (example, continued)

- **However**, one pole known to be exactly at the origin:

$$H^0(s) = \frac{1}{s}$$

  Term $H^0(j\omega)$ is shown in table 5.

- The influence of the $H^0(s)$ term is removed from $\overline{H}(j\omega)$, to give $\overline{H}^1(j\omega)$, also seen in table 5.

- When a 1st order model is estimated from $\overline{H}^1(j\omega)$ with invfreqs(), the result is:

$$\widehat{H}^1(s) = \frac{24.30}{1.0s + 3.97} \tag{28}$$

- Putting the known dynamics back in gives:

$$\widehat{H}(s) = H^0(s)\,\widehat{H}^1(s) = \frac{1}{s}\,\frac{24.30}{1.0s + 3.97} \tag{29}$$

  which matches $H^*(s)$ substantially better than does $\widehat{H}(s)$ given directly from the data.

	1st pole location	2nd pole location	DC Gain (3)
True	$s = 0$	$s = -4$	6.0
Full Model Estimation	$s = -0.023$	$s = -5.06$	$\approx \frac{24.3}{5.06} = 4.8$
Using known dynamics	$s = 0$	$s = -3.97$	6.12

Table 6: Characteristics of $H^*(s)$ and two estimates $\widehat{H}(s)$.

Notes: (3) DC Gain as a special meaning for systems with one or more poles at the origin; which we will see in Lecture Notes 16.

3.3.9 Conclusions, System ID from frequency response

- Using frequency response, modes beyond the dominant mode can be identified.
- If *a priori* information is available: **use it.**
  - If the model structure is known (say, from first-principles analysis of the system), use the known model structure.
  - Augment the data, for example if the DC gain is known.
  - Reduce the order of the model to be estimated, if some poles or zeros are known.
- Estimating a low-order model is easier and more accurate than estimating a high-order model.
- System Identification is not a blind, turn-the-crank type process. The Controls Engineer's knowledge of the system being identified must be used to verify the correctness of the results each step of the way.

4 Conclusions

- Simulation is used to study non-linear systems.
- System Identification is the process of determining a system model from experimental data.
  - By step response;
  - By frequency response;
  - By advanced methods.
- System ID has two parts:
  1. Determining the model structure;
  2. Determining values of parameters within the model structure.
- The System ID problem is quite common, the system model is often not completely known from other information.
- Use *a priori* information whenever it is available.
  - Improved accuracy;
  - Resulting model meets physical constraints.